

Wage Inequality, Upskilling, and Long-Run Growth: A Three-Player Model

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Abstract

We develop a tractable three-player model connecting wage inequality, worker upskilling decisions, and firm asset accumulation. A single parameter $\mu \in (0, \frac{1}{2})$ — the income share of the lower-ranked worker group — governs the entire distribution of wages. Workers allocate income between upskilling investment and participation in a zero-sum displacement lottery. Preferences are modelled using linear utility augmented by a signed curvature parameter θ_s per worker: $\theta_L > 0$ for the lower-income worker (risk-seeking) and $\theta_H < 0$ for the higher-income worker (risk-averse). This two-parameter specification replaces the full Friedman-Savage machinery while preserving the essential risk asymmetry. All equilibrium quantities — upskilling fractions, lottery stakes, firm asset accumulation, and steady-state wage split — remain in closed form. We distinguish two upskilling measures: an individual measure $K_{\text{ind}}^* = 2\kappa^{\min}(\mu)$ independent of population structure, and an aggregate measure $K_{\text{agg}}^* = (W + 1)\kappa^{\min}(\mu)$ weighted by the population mass ratio $W = i^*/(N - i^*)$, which enters firm growth through the upskilling index $\Phi(\mu)$. We address two questions: how do workers respond to wage inequality μ , and what constraints does μ place on long-term growth?

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1 Introduction

How should a firm set wages across skill groups to maximise long-run asset growth? And how do workers in different income groups respond to the wage structure they face? This paper provides a unified framework with closed-form answers to both questions.

There are three players: a lower-income worker L , a higher-income worker H , and a firm. The firm holds productive assets and pays wages as a share of those assets. It chooses a wage-split parameter $\mu \in (0, \frac{1}{2})$ each period — the income share of the lower-income group — trading off the upskilling return from greater equality against a convex restructuring cost. Workers receive wages determined by μ and split income between upskilling investment and participation in a zero-sum displacement lottery. A participation constraint ensures each worker invests enough to maintain a subsistence base income.

Preferences are modelled parsimoniously. The baseline is linear utility $u(m) = m$, augmented by a single curvature correction per worker:

$$u_s(m) = m + \frac{\theta_s}{2}(m - \bar{b})^2 \quad (1)$$

where $\bar{b}$ is the subsistence income and $\theta_s = u''_s(\bar{b})$ is the signed curvature at subsistence. Setting $\theta_L > 0$ makes worker L locally risk-seeking (convex utility) and $\theta_H < 0$ makes worker H locally risk-averse (concave utility). This replaces the three-parameter Friedman-Savage specification (α_u, β_u, m^*) with two parameters (θ_L, θ_H) , while retaining the key qualitative asymmetry. Crucially, these parameters affect only the welfare evaluation of lottery participation, not the equilibrium allocation quantities, which remain unchanged from the linear case.

A central structural result concerns the role of the population mass ratio $W = i^*/(N - i^*)$, where i^* is the class barrier separating lower- and upper-income workers in the underlying N -worker model. At the individual level, each representative worker's upskilling investment $\kappa^{\min}(\mu)$ is determined solely by the participation constraint, with no dependence on W . At the economy-wide level, however, the population-weighted aggregate $(W + 1)\kappa^{\min}(\mu)$ does depend on W and is the quantity the firm internalises through its upskilling index $\Phi(\mu)$. This distinction — W enters growth through aggregation, not through individual incentives — is a main structural result of the model.

2 Model Setup

2.1 Players and Incomes

There are three strategic players: worker L , worker H , and a firm. The firm holds asset stock A_t and distributes total income R to workers. It chooses the income-split parameter $\mu \in (0, \frac{1}{2})$:

$$M_L = R\mu, \quad M_H = R(1 - \mu) \quad (2)$$

Since $\mu < \frac{1}{2}$, worker H always earns more than worker L . Higher μ means greater wage equality; lower μ means greater inequality. The income ratio $M_H/M_L = (1 - \mu)/\mu$ is strictly decreasing in μ .

Definition 1 (Wage inequality parameter). $\mu \in (0, \frac{1}{2})$ is the income share of worker L , chosen by the firm each period. It is the single measure of wage inequality in the model.

In the underlying N -worker model with a Pareto income ladder $m_i = x_0(1 - i/(N+1))^{-1/\alpha}$ and a class barrier at rank i^* , the two-player reduction yields μ as the lower hump's share of total representative income:

$$\mu \approx \frac{\int_0^{i^*/N} x_0(1 - s)^{-1/\alpha} ds}{\int_0^1 x_0(1 - s)^{-1/\alpha} ds} \quad (3)$$

In the macroeconomic model μ is taken as the firm's policy instrument rather than a fixed structural outcome.

2.2 The Population Mass Ratio W

The population mass ratio is:

$$W = \frac{i^*}{N - i^*} \geq 1 \quad (4)$$

Under the Pareto distribution the lower group is generically more numerous ($W \geq 1$). The parameter W enters the model in two distinct and separate ways:

1. **Lottery pool:** W amplifies worker L 's aggregate stake in the prize pot, reflecting the greater frequency of lower-rank displacement events per period.
2. **Aggregate upskilling:** W weights the lower group's upskilling in the population-weighted total $(W + 1)\kappa^{\min}$, which the firm uses to compute the return on its wage bill.

W does *not* affect any individual worker's budget or participation constraint. Each representative worker invests $\kappa^{\min}(\mu)$ regardless of W .

2.3 Worker Budget and Allocation

Each worker $s \in \{L, H\}$ allocates fraction $\lambda_s \in [0, 1]$ of income M_s to upskilling and the remainder to the lottery:

$$\kappa_s = \lambda_s M_s \quad (\text{upskilling}), \quad \ell_s = (1 - \lambda_s) M_s \quad (\text{lottery stake}) \quad (5)$$

Upskilling produces output $f(\kappa_s) = \kappa_s^\gamma$, $\gamma \in (0, 1)$, with diminishing returns. The certain base income after upskilling and human capital depreciation $\delta_w > 0$:

$$b_s(\lambda_s) = \kappa_s + \kappa_s^\gamma - \delta_w \quad (6)$$

2.4 Two Upskilling Measures

Definition 2 (Individual upskilling total). $K_{\text{ind}}^* = \kappa_L^* + \kappa_H^* = 2\kappa^{\min}(\mu)$ is the sum of the two representative agents' equilibrium investments. It is independent of W .

Definition 3 (Aggregate upskilling total). $K_{\text{agg}}^* = W\kappa_L^* + \kappa_H^* = (W + 1)\kappa^{\min}(\mu)$ is the population-weighted sum. It enters the firm's asset accumulation through the upskilling index $\Phi(\mu) = K_{\text{agg}}^*/R$.

The ratio $K_{\text{agg}}^*/K_{\text{ind}}^* = (W + 1)/2 \geq 1$. For $W > 1$, aggregate upskilling exceeds the two-agent sum because the more numerous lower group amplifies the economy-wide human capital base.

2.5 Zero-Sum Displacement Lottery

The lottery pool is:

$$\Lambda = W(1 - \lambda_L)M_L + (1 - \lambda_H)M_H = W\ell_L + \ell_H \quad (7)$$

Each worker wins the pool with probability $\frac{1}{2}$. Outcome incomes are:

$$M_L^{\text{win}} = b_L + (1 - \lambda_H)M_H \quad (8)$$

$$M_L^{\text{lose}} = b_L - W(1 - \lambda_L)M_L \quad (9)$$

$$M_H^{\text{win}} = b_H + W(1 - \lambda_L)M_L \quad (10)$$

$$M_H^{\text{lose}} = b_H - (1 - \lambda_H)M_H \quad (11)$$

The lottery is strictly zero-sum: Λ is redistributed, not created. Only upskilling changes total income.

2.6 Participation Constraint

Each worker must maintain base income above a subsistence floor $\bar{b}(\mu) = \bar{b}_0 R\mu$, proportional to the lower worker's wage:

$$b_s(\lambda_s) \geq \bar{b}(\mu) \iff \kappa_s + \kappa_s^\gamma \geq \bar{b}_0 R\mu + \delta_w \quad (12)$$

The minimum upskilling satisfying this constraint, $\kappa^{\min}(\mu)$, solves:

$$\kappa^{\min} + (\kappa^{\min})^\gamma = \bar{b}_0 R\mu + \delta_w \quad (13)$$

For $\gamma = 2$, adopted throughout for closed form:

$$\kappa^{\min}(\mu) = \frac{-1 + \sqrt{1 + 4(\bar{b}_0 R\mu + \delta_w)}}{2} \quad (14)$$

This is strictly increasing and concave in μ , with sensitivity:

$$\frac{d\kappa^{\min}}{d\mu} \equiv \Psi(\mu) = \frac{\bar{b}_0 R}{1 + 2\kappa^{\min}(\mu)} > 0 \quad (15)$$

3 Preferences: Linear Utility with Curvature Correction

3.1 The Two-Parameter Specification

We use linear utility augmented by a quadratic curvature correction evaluated at the subsistence income $\bar{b}$:

$$u_s(m) = m + \frac{\theta_s}{2}(m - \bar{b})^2 \quad (16)$$

where $\theta_s \in \mathbb{R}$ is the curvature parameter for worker s . The derivatives are:

$$u'_s(m) = 1 + \theta_s(m - \bar{b}) \quad (17)$$

$$u''_s(m) = \theta_s \quad (18)$$

The curvature $u''_s = \theta_s$ is constant — the simplest possible departure from linearity. We impose:

$$\theta_L > 0 \quad (\text{worker } L: \text{convex utility, risk-seeking}), \quad \theta_H < 0 \quad (\text{worker } H: \text{concave utility, risk-averse}) \quad (19)$$

This two-parameter specification directly encodes the Friedman-Savage risk asymmetry without requiring an inflection point or a polynomial of degree greater than two.

Remark 1. Equation (16) is equivalent to a second-order Taylor expansion of any smooth utility function around $\bar{b}$, retaining only the zeroth, first, and second-order terms. The parameter $\theta_s = u''_s(\bar{b})$ is the Arrow-Pratt numerator evaluated at subsistence. The Arrow-Pratt denominator $u'_s(\bar{b}) = 1$ under this specification, giving absolute risk aversion $A_s(\bar{b}) = -\theta_s$.

3.2 Risk Premia

For a binary lottery paying $\bar{b} + x$ or $\bar{b} - y$ with equal probability, the expected utility under specification (16) is:

$$\frac{1}{2}u_s(\bar{b} + x) + \frac{1}{2}u_s(\bar{b} - y) = \bar{b} + \frac{x - y}{2} + \frac{\theta_s}{2} \cdot \frac{x^2 + y^2}{2} \quad (20)$$

The first two terms are the linear expected value. The third term $\frac{\theta_s}{4}(x^2 + y^2)$ is the risk premium correction. Using the compact notation:

$$a(\mu) = R\mu - \kappa^{\min}(\mu) \quad (\text{worker } L \text{ stake}) \quad (21)$$

$$c(\mu) = R(1 - \mu) - \kappa^{\min}(\mu) \quad (\text{worker } H \text{ stake}) \quad (22)$$

$$\Pi(\mu) = Wa(\mu) + c(\mu) \quad (\text{total pool spread}) \quad (23)$$

$$\epsilon_L(\mu) = \frac{c(\mu) - Wa(\mu)}{2} \quad (\text{net expected transfer to } L) \quad (24)$$

$$\epsilon_H(\mu) = -\epsilon_L(\mu) \quad (\text{net expected transfer to } H) \quad (25)$$

the win and lose amounts for worker s are $x_s = \Pi_{-s}$ (prize received) and $y_s = W_s \ell_s$ (stake lost). Substituting into equation (20):

$$V_s = \bar{b} + \epsilon_s + \underbrace{\frac{\theta_s}{4}(\Pi_{-s}^2 + (W_s \ell_s)^2)}_{\tilde{\rho}_s} \quad (26)$$

At the constrained equilibrium where $b_s = \bar{b}$, $\Pi_L = c$, $\Pi_H = Wa$, $W_L \ell_L = Wa$, $W_H \ell_H = c$:

$$\tilde{\rho}_L = \frac{\theta_L}{4}(c^2 + W^2 a^2), \quad \tilde{\rho}_H = \frac{\theta_H}{4}(W^2 a^2 + c^2) \quad (27)$$

Writing $\Sigma^2 \equiv c^2 + W^2 a^2$, the risk premia take the symmetric form:

$$\boxed{\tilde{\rho}_s = \frac{\theta_s}{4}\Sigma^2} \quad (28)$$

Since $\theta_L > 0$: $\tilde{\rho}_L > 0$ — worker L gains welfare from the lottery beyond expected income.
 Since $\theta_H < 0$: $\tilde{\rho}_H < 0$ — worker H loses welfare from the lottery.

The net social risk premium is:

$$\tilde{\rho}_L + \tilde{\rho}_H = \frac{\theta_L + \theta_H}{4} \Sigma^2 \quad (29)$$

The lottery displacement mechanism is socially desirable if and only if:

$$\theta_L + \theta_H > 0 \quad (30)$$

The lower worker's risk-seeking gain outweighs the upper worker's risk-aversion loss. This is a single inequality in two observable parameters, requiring no reference to an inflection point.

3.3 Equilibrium Utility

At the constrained equilibrium ($b_s = \bar{b}$), the full utility is:

$$V_s = \underbrace{\bar{b}}_{\text{subsistence}} + \underbrace{\epsilon_s}_{\text{expected transfer}} + \underbrace{\frac{\theta_s}{4} \Sigma^2}_{\text{risk premium}} \quad (31)$$

The three terms are cleanly separated. The curvature parameters (θ_L, θ_H) affect only the third term. They do not enter the participation constraint, the optimal λ_s^* , the firm's FOC, or any accumulation object. This clean separation is the main advantage of the quadratic specification: risk preferences are welfare relevant but allocation neutral.

4 Derivation of Optimal λ_s^*

4.1 Why the FOC Has No Interior Solution

Worker s maximises $V_s(\lambda_s; \lambda_{-s})$ subject to $b_s(\lambda_s) \geq \bar{b}$. Differentiating with respect to λ_s :

$$\frac{\partial V_s}{\partial \lambda_s} = \frac{\partial b_s}{\partial \lambda_s} + \frac{\partial \epsilon_s}{\partial \lambda_s} + \frac{\theta_s}{4} \frac{\partial \Sigma^2}{\partial \lambda_s} \quad (32)$$

Using $\partial b_s / \partial \lambda_s = M_s(1 + \gamma \kappa_s^{\gamma-1}) > 0$, $\partial \epsilon_s / \partial \lambda_s$ from the stake expressions, and $\partial \Sigma^2 / \partial \lambda_s$, each component is bounded and finite. However, the dominant term is $\partial b_s / \partial \lambda_s > 0$: raising λ_s raises base income in both outcome states, and for small θ_s this dominates. More precisely, taking the full derivative and examining signs shows $\partial V_s / \partial \lambda_s > 0$ over the relevant range, so V_s is increasing in λ_s . There is no interior zero of the FOC.

The economic reason is that raising λ_s improves both the win outcome ($b_s + \Pi_{-s}$ rises as b_s rises) and the lose outcome ($b_s - W_s \ell_s$ rises as b_s rises and ℓ_s falls). Both states improve simultaneously, so the unconstrained optimum is the corner $\lambda_s = 1$.

4.2 The Binding Participation Constraint

Since V_s is increasing in λ_s and the worker cannot set $\lambda_s = 1$ without violating the subsistence floor (at $\lambda_s = 1$, $\kappa_s = M_s$ but b_s may exceed $\bar{b}$ — the constraint binds from below as λ_s is pushed up until b_s just reaches $\bar{b}$), the optimal solution is:

$$b_s(\lambda_s^*) = \bar{b} \iff \kappa_s^* + (\kappa_s^*)^\gamma = \bar{b} + \delta_w \quad (33)$$

This is precisely the equation defining $\kappa^{\min}$. Since $\kappa_s^* = \lambda_s^* M_s = \lambda_s^* R M_s^0$:

$$\lambda_s^* = \frac{\kappa^{\min}(\mu)}{R M_s^0} \quad (34)$$

For worker L ($M_L^0 = \mu$) and worker H ($M_H^0 = 1 - \mu$):

$$\boxed{\lambda_L^*(\mu) = \frac{\kappa^{\min}(\mu)}{R\mu}, \quad \lambda_H^*(\mu) = \frac{\kappa^{\min}(\mu)}{R(1 - \mu)}} \quad (35)$$

4.3 Ordering and Separation

Since $R\mu < R(1 - \mu)$, the ordering $\lambda_L^* > \lambda_H^*$ holds always. The lower-income worker invests a larger fraction in upskilling because the same absolute floor $\kappa^{\min}$ represents a larger share of a smaller budget. This is an income-scaling effect, not a preference effect. The curvature parameters (θ_L, θ_H) do not appear in equation (35).

5 Question 1: Worker Responses to Wage Inequality

μ

5.1 Upskilling Responses

Differentiating equation (35):

$$\frac{d\lambda_L^*}{d\mu} = \frac{\Psi(\mu) - \kappa^{\min}(\mu)/\mu}{R\mu} \quad (36)$$

$$\frac{d\lambda_H^*}{d\mu} = \frac{\Psi(\mu) + \kappa^{\min}(\mu)/(1 - \mu)}{R(1 - \mu)} > 0 \quad \text{always} \quad (37)$$

Worker H always raises their upskilling fraction when equality rises because both terms in the numerator of (37) are positive. For worker L , the sign depends on whether $\Psi(\mu) > \kappa^{\min}(\mu)/\mu$, which holds when the subsistence requirement is tight relative to current investment.

The two upskilling totals and their responses to μ :

$$K_{\text{ind}}^*(\mu) = 2\kappa^{\min}(\mu), \quad \frac{dK_{\text{ind}}^*}{d\mu} = 2\Psi(\mu) > 0 \quad (38)$$

$$K_{\text{agg}}^*(\mu) = (W+1)\kappa^{\min}(\mu), \quad \frac{dK_{\text{agg}}^*}{d\mu} = (W+1)\Psi(\mu) > 0 \quad (39)$$

The aggregate measure responds more strongly to μ by the factor $(W+1)/2$:

$$\frac{dK_{\text{agg}}^*/d\mu}{dK_{\text{ind}}^*/d\mu} = \frac{W+1}{2} \quad (40)$$

Proposition 1 (Two-level upskilling neutrality). *At the constrained equilibrium:*

1. $K_{\text{ind}}^* = 2\kappa^{\min}(\mu)$ is independent of W . Redistribution holding μ fixed does not alter the sum of individual investments.
2. $K_{\text{agg}}^* = (W+1)\kappa^{\min}(\mu)$ is increasing in W . A larger lower-rank population amplifies the economy-wide human capital base.
3. Only K_{agg}^* enters firm growth through $\Phi(\mu) = K_{\text{agg}}^*/R$.

5.2 Lottery Stakes, Net Transfer, and Risk Premia

The lottery stakes respond to μ as:

$$\frac{da}{d\mu} = R - \Psi(\mu) > 0, \quad \frac{dc}{d\mu} = -R - \Psi(\mu) < 0 \quad (41)$$

Worker L 's stake rises (income rises faster than the floor); worker H 's stake falls (income falls and floor rises). The net transfer to L :

$$\epsilon_L(\mu) = \frac{c(\mu) - Wa(\mu)}{2} \quad (42)$$

is decreasing in μ when $W > 1$: higher equality reduces worker L 's net lottery receipt because their larger stake means they contribute more to the pool.

The risk premium $\tilde{\rho}_s = \theta_s \Sigma^2/4$ responds to μ through $\Sigma^2 = c^2 + W^2 a^2$:

$$\frac{d\Sigma^2}{d\mu} = 2c \frac{dc}{d\mu} + 2W^2 a \frac{da}{d\mu} = -2c(R + \Psi) + 2W^2 a(R - \Psi) \quad (43)$$

whose sign depends on the relative sizes of the stakes and W . For the social premium, since $\theta_L + \theta_H$ is a fixed constant, the sign of $d(\tilde{\rho}_L + \tilde{\rho}_H)/d\mu$ equals the sign of $d\Sigma^2/d\mu$. Greater equality can increase or decrease the net social welfare from the lottery depending

on whether the gain in worker L 's stake dominates or is dominated by the loss in worker H 's stake.

- Proposition 2** (Worker responses to wage equality). *1. Worker H always raises λ_H^* when μ increases. Worker L raises λ_L^* when the participation constraint is sufficiently tight.*
- 2. K_{ind}^* is increasing in μ and independent of W .*
- 3. K_{agg}^* is increasing in both μ and W , with amplification factor $(W + 1)/2$.*
- 4. The lottery is socially desirable iff $\theta_L + \theta_H > 0$, a condition independent of μ .*

5.3 Summary of Worker Responses

Table 1 collects all signed responses.

Table 1: Worker responses to an increase in wage equality ($\mu \uparrow$)

Object	Symbol	$\mu \uparrow$	Mechanism
Worker L income	M_L	+	Direct
Worker H income	M_H	−	Direct
Upskilling floor	$\kappa^{\min}$	+	Rising $\bar{b}_0 R \mu$
Worker L upskill fraction	λ_L^*	+	(cond.) Eq. (36)
Worker H upskill fraction	λ_H^*	+	(always) Both terms positive
Individual total	$K_{\text{ind}}^* = 2\kappa^{\min}$	+	Floor rises; W -free
Aggregate total	$K_{\text{agg}}^* = (W+1)\kappa^{\min}$	+	Floor rises; W -amplified
Worker L stake	a	+	Income outruns floor
Worker H stake	c	−	Income falls, floor rises
Net transfer to L	ϵ_L^*	−	Larger L stake hurts L
L risk premium	$\tilde{\rho}_L = \theta_L \Sigma^2/4$	amb.	Via Σ^2
H risk premium	$\tilde{\rho}_H = \theta_H \Sigma^2/4$	amb.	Via Σ^2
Net social premium	$(\theta_L + \theta_H) \Sigma^2/4$	amb.	Sign fixed by $\theta_L + \theta_H$

6 Question 2: μ Constraints on Long-Term Growth

6.1 The Firm's Asset Accumulation Problem

The firm accumulates assets A_t according to:

$$A_{t+1} = [(1 - \pi - \delta_f) + \pi \Phi(\mu_t)] A_t - K(\mu_t) - C(A_t) \quad (44)$$

where:

$$\Phi(\mu_t) = \frac{K_{\text{agg}}^*(\mu_t)}{R} = \frac{(W+1)\kappa^{\min}(\mu_t)}{R} \quad (45)$$

$$K(\mu_t) = k|\mu_t - \mu_0|^\eta, \quad k, \eta > 0 \quad (46)$$

$$C(A_t) = c_f A_t^\xi, \quad c_f, \xi > 0 \quad (47)$$

The upskilling index $\Phi(\mu_t)$ is the aggregate upskilling per unit of total income. The connection between the two upskilling measures is:

$$\Phi(\mu) = \frac{K_{\text{agg}}^*}{R} = \frac{W+1}{2} \cdot \frac{K_{\text{ind}}^*}{R} \quad (48)$$

When $W = 1$ the two measures coincide. For $W > 1$ the firm's return to raising μ exceeds what a two-agent welfare analysis alone suggests.

6.2 Stackelberg Timing

The firm moves first by setting μ_t , anticipating workers' best-response functions $\lambda_s^*(\mu_t)$ from equation (35). By substituting these before optimising, the firm's problem reduces to a single-variable maximisation with no circularity:

$$\mu_t^* = \arg \max_{\mu_t} \left\{ \left[(1 - \pi - \delta_f) + \frac{\pi(W+1)\kappa^{\min}(\mu_t)}{R} \right] A_t - k|\mu_t - \mu_0|^\eta - c_f A_t^\xi \right\} \quad (49)$$

6.3 Firm's First-Order Condition and Optimal μ^*

With $\eta = 2$ and Ψ evaluated at reference $\Psi_0 \equiv \Psi(\mu_0)$:

$$\underbrace{\frac{\pi(W+1)\Psi_0}{R} \cdot A_t}_{\text{marginal upskilling gain}} = \underbrace{2k(\mu_t^* - \mu_0)}_{\text{marginal restructuring cost}} \quad (50)$$

Solving:

$$\boxed{\mu_t^* = \mu_0 + \Gamma A_t, \quad \Gamma \equiv \frac{\pi(W+1)\Psi_0}{2kR}} \quad (51)$$

The coefficient Γ is proportional to $(W+1)$: the firm's incentive to equalise wages is driven by the aggregate upskilling return, not individual worker responses.

Proposition 3 (Firm's optimal wage inequality). *1. $\mu_t^* = \mu_0 + \Gamma A_t$: optimal equality is linear in asset wealth.*

2. $\mu_t^ > \mu_0$ for all $A_t > 0$: the firm always chooses greater equality than the market reference.*

3. $\partial\mu_t^*/\partial A_t = \Gamma > 0$: wealthier firms enforce greater equality.
4. $\partial\mu_t^*/\partial W > 0$: larger lower-rank population increases the optimal equality through $(W + 1)$ in Γ .
5. $\partial\mu_t^*/\partial k < 0$: higher restructuring costs compress the deviation from μ_0 .

6.4 Three Constraints of μ on Long-Term Growth

The asset growth rate is:

$$g(\mu, A) = \pi\Phi(\mu) - \delta_f - c_f - \frac{k(\mu - \mu_0)^2}{A} \quad (52)$$

Constraint 1 — Viability floor. For a positive steady state to exist, evaluating at $\mu = \mu_0$ (no restructuring cost) and $A \rightarrow \infty$:

$$\boxed{\pi(W + 1)\kappa_0 > R(\delta_f + c_f)} \quad (53)$$

where $\kappa_0 = \kappa^{\min}(\mu_0)$. This is the fundamental constraint: the aggregate upskilling return must exceed total running costs. A larger lower-rank group (W higher) makes viability easier to satisfy. When this fails, no wage policy can generate positive steady-state growth.

Constraint 2 — Restructuring upper bound. For the firm to benefit from $\mu_t > \mu_0$:

$$\pi[\Phi(\mu_t) - \Phi(\mu_0)]A_t > k(\mu_t - \mu_0)^2 \quad (54)$$

Richer firms can sustain greater deviations from μ_0 . As $A_t \rightarrow A^*$ this constraint binds at $\mu_t = \mu_t^*$ from equation (51).

Constraint 3 — Hard feasibility ceiling. $\mu < \frac{1}{2}$ always: worker H must earn more than worker L . As $\mu \rightarrow \frac{1}{2}$ the upskilling gap closes and the lottery spread $\Pi(\mu) \rightarrow 0$.

6.5 Closed-Form Steady-State Equilibrium

Setting $A_{t+1} = A_t = A^*$ and using $\mu^* - \mu_0 = \Gamma A^*$:

$$A^*[\pi\Phi^* - \delta_f - c_f] = k\Gamma^2 A^{*2} \quad (55)$$

Dividing by $A^* > 0$, evaluating Φ^* at κ_0 :

$$\boxed{A^* = \frac{\pi(W + 1)\kappa_0/R - \delta_f - c_f}{k\Gamma^2}} \quad (56)$$

Define three derived constants computed once from primitives:

$$\kappa_0 = \frac{-1 + \sqrt{1 + 4(\bar{b}_0 R \mu_0 + \delta_w)}}{2} \quad (57)$$

$$\Psi_0 = \frac{\bar{b}_0 R}{1 + 2\kappa_0} \quad (58)$$

$$\Gamma = \frac{\pi(W + 1)\Psi_0}{2kR} \quad (59)$$

All steady-state objects follow sequentially:

$$\text{Firm:} \quad A^* = \frac{\pi(W + 1)\kappa_0/R - \delta_f - c_f}{k\Gamma^2} \quad (60)$$

$$\mu^* = \mu_0 + \Gamma A^* \quad (61)$$

$$\text{Workers:} \quad \kappa^* = \frac{-1 + \sqrt{1 + 4(\bar{b}_0 R \mu^* + \delta_w)}}{2} \quad (62)$$

$$\lambda_L^* = \frac{\kappa^*}{R\mu^*}, \quad \lambda_H^* = \frac{\kappa^*}{R(1 - \mu^*)} \quad (63)$$

$$K_{\text{ind}}^* = 2\kappa^*, \quad K_{\text{agg}}^* = (W + 1)\kappa^* \quad (64)$$

$$a^* = R\mu^* - \kappa^*, \quad c^* = R(1 - \mu^*) - \kappa^* \quad (65)$$

$$\Lambda^* = W a^* + c^*, \quad \epsilon_L^* = \frac{c^* - W a^*}{2} \quad (66)$$

$$\text{Risk premia:} \quad \tilde{\rho}_s^* = \frac{\theta_s}{4} (c^{*2} + W^2 a^{*2}) \quad (67)$$

Every object is an explicit function of the primitives $(R, \mu_0, W, \bar{b}_0, \delta_w, \pi, \delta_f, k, c_f, \theta_L, \theta_H)$. Note that (θ_L, θ_H) appear only in (67) and nowhere in equations (60)–(66).

6.6 The Endogenous Kuznets Relationship

At the steady state $\mu^* = \mu_0 + \Gamma A^*$, so wealthier economies operate at a more equal wage split. Along the transitional path, as A_t grows from A_0 toward A^* , $\mu_t^* = \mu_0 + \Gamma A_t$ rises monotonically. The income inequality ratio $(1 - \mu^*)/\mu^*$ falls as the economy grows: an endogenous Kuznets relationship in which growth and equality are complementary, driven by the firm's optimal wage policy.

7 Full Comparative Statics

Table 2 presents the complete steady-state comparative statics. The W row is the most important: K_{ind}^* is invariant to W (shown as **0**) while K_{agg}^* rises with W (shown as +). The risk premium parameters (θ_L, θ_H) appear only in the last column.

Table 2: Steady-state comparative statics for all three players

Shock ($\uparrow$)	∂A^*	$\partial \mu^*$	$\partial \lambda_L^*$	$\partial \lambda_H^*$	$\partial K_{\text{ind}}^*$	$\partial K_{\text{agg}}^*$
$W \uparrow$	+	+	amb.	—	0	+
$\pi \uparrow$	+	+	+	+	+	+
$k \uparrow$	—	—	—	—	—	—
$\bar{b}_0 \uparrow$	amb.	+	+	—	+	+
$\delta_f \uparrow$	—	—	—	—	—	—
$\delta_w \uparrow$	+	+	+	amb.	+	+
$\mu_0 \uparrow$	+	+	amb.	—	+	+
$R \uparrow$	amb.	amb.	—	—	amb.	amb.
$\theta_L \uparrow$	—	—	—	—	—	—
$\theta_H \uparrow$	—	—	—	—	—	—

Notes: “amb.” = ambiguous sign. “—” = parameter does not enter the allocation or accumulation problem. **0** denotes exact invariance. (θ_L, θ_H) affect only $\tilde{\rho}_s^* = \theta_s(c^{*2} + W^2 a^{*2})/4$: sign of $\partial \tilde{\rho}_s / \partial \theta_s > 0$, sign of $\partial(\tilde{\rho}_L + \tilde{\rho}_H) / \partial(\theta_L + \theta_H) > 0$.

The (θ_L, θ_H) rows confirm the separation property: risk preference parameters have no effect on any allocation or accumulation quantity. They affect only the welfare valuation of lottery participation through $\tilde{\rho}_s^*$.

8 Conclusion

We have developed a three-player model in which wage inequality μ connects worker behaviour to firm growth, with all equilibrium objects in closed form. Risk preferences are captured by two signed curvature parameters $(\theta_L > 0, \theta_H < 0)$ that augment linear utility, replacing the three-parameter Friedman-Savage specification while retaining the essential asymmetry: the lower-income worker gains welfare from the lottery and the higher-income worker loses.

On the first question — how workers respond to μ — the main findings are: worker H always raises their upskilling fraction when equality rises; individual total upskilling $K_{\text{ind}}^* = 2\kappa^{\min}(\mu)$ rises with μ but is invariant to W ; aggregate total upskilling $K_{\text{agg}}^* = (W + 1)\kappa^{\min}(\mu)$ rises with both μ and W by amplification factor $(W + 1)/2$; and the lottery is socially desirable iff $\theta_L + \theta_H > 0$.

On the second question — constraints of μ on growth — the main findings are: the viability condition $\pi(W + 1)\kappa_0 > R(\delta_f + c_f)$ is the growth floor; conditional on viability the firm’s optimal $\mu^* = \mu_0 + \Gamma A^*$ is linear in wealth, producing an endogenous Kuznets relationship; and A^* is fully explicit in nine primitives with risk parameters entering only

through the welfare evaluation of the lottery.

A Complete Notation Table

A Derivation of Optimal λ_s^*

B.1 The Worker's Problem

Worker s maximises V_s from equation (26) subject to $b_s(\lambda_s) \geq \bar{b}$. The objective is:

$$V_s = \bar{b} + \epsilon_s(\lambda_s, \lambda_{-s}) + \frac{\theta_s}{4} \Sigma^2(\lambda_s, \lambda_{-s}) \quad (\text{B.1})$$

B.2 Monotonicity of V_s in λ_s

Differentiating with respect to λ_s at the constrained equilibrium $b_s = \bar{b}$:

$$\frac{\partial V_s}{\partial \lambda_s} = \frac{\partial \epsilon_s}{\partial \lambda_s} + \frac{\theta_s}{4} \frac{\partial \Sigma^2}{\partial \lambda_s} \quad (\text{B.2})$$

Since $\epsilon_s = (c - W_s \ell_s - \Pi_{-s})/2$ evaluated at stakes and $\Sigma^2 = \Pi_{-s}^2 + W_s^2 \ell_s^2$, both terms have contributions that, when combined with the base income channel $\partial b_s / \partial \lambda_s = M_s(1 + \gamma \kappa_s^{\gamma-1}) > 0$, make $\partial V_s / \partial \lambda_s > 0$ throughout the feasible range. This is because raising λ_s simultaneously improves both the win outcome (through higher b_s) and the lose outcome (through higher b_s and lower stake ℓ_s). There is no interior zero.

B.3 The Constraint Binds

Since V_s is increasing in λ_s , the worker pushes λ_s as high as possible. The participation constraint $b_s \geq \bar{b}$ prevents $\lambda_s \rightarrow 1$ (which would leave zero lottery stake but may still satisfy the floor). The binding condition is:

$$\kappa_s^* + (\kappa_s^*)^\gamma = \bar{b} + \delta_w \implies \kappa^{\min}(\mu) = \frac{-1 + \sqrt{1 + 4(\bar{b} + \delta_w)}}{2} \quad [\gamma = 2] \quad (\text{B.3})$$

Since $\kappa_s^* = \lambda_s^* M_s$:

$$\lambda_s^* = \frac{\kappa^{\min}(\mu)}{M_s} = \frac{\kappa^{\min}(\mu)}{RM_s^0} \quad (\text{B.4})$$

B.4 Ordering and Separation of Risk Preferences

The ordering $\lambda_L^* > \lambda_H^*$ follows from $R\mu < R(1 - \mu)$. This is an income-scaling effect: the same floor $\kappa^{\min}$ represents a larger fraction of a smaller income.

The curvature parameters (θ_L, θ_H) do not appear in $\kappa^{\min}(\mu)$ or in λ_s^* . They enter only the risk premium $\tilde{\rho}_s = \theta_s \Sigma^2 / 4$ in the utility evaluation (31). The allocation problem and the preference problem are completely separated.

Table 3: All symbols, definitions, and domains

Symbol	Definition	Domain
<i>Primitives</i>		
R	Total income to workers	$\mathbb{R}_{++}$
μ	Income share of worker L	$(0, \frac{1}{2})$
μ_0	Reference (market) income split	$(0, \frac{1}{2})$
W	Population mass ratio = $i^*/(N - i^*)$	$[1, \infty)$
$\bar{b}_0$	Subsistence coefficient; $\bar{b} = \bar{b}_0 R \mu$	$\mathbb{R}_{++}$
δ_w	Human capital depreciation	$[0, 1)$
γ	Upskilling power (= 2 for closed form)	$(0, 1)$
π	Firm wage share of assets	$(0, 1)$
δ_f	Firm asset depreciation	$(0, 1)$
k	Restructuring cost scale	$\mathbb{R}_{++}$
η	Restructuring cost power (= 2 for closed form)	$(1, \infty)$
c_f	Maintenance cost scale	$\mathbb{R}_{++}$
ξ	Maintenance cost power	$(1, \infty)$
θ_L	Curvature of u_L at $\bar{b}$; $\theta_L > 0$	$\mathbb{R}_{++}$
θ_H	Curvature of u_H at $\bar{b}$; $\theta_H < 0$	$\mathbb{R}_{--}$
<i>Derived constants</i>		
κ_0	$\kappa^{\min}(\mu_0) = (-1 + \sqrt{1 + 4(\bar{b}_0 R \mu_0 + \delta_w)})/2$	$\mathbb{R}_{++}$
Ψ_0	Sensitivity at μ_0 : $\bar{b}_0 R / (1 + 2\kappa_0)$	$\mathbb{R}_{++}$
Γ	Firm FOC coefficient: $\pi(W + 1)\Psi_0 / (2kR)$	$\mathbb{R}_{++}$
<i>Worker objects</i>		
M_L, M_H	Incomes = $R\mu, R(1 - \mu)$	$\mathbb{R}_{++}$
λ_s	Fraction of income to upskilling	$[0, 1]$
κ_s	Upskilling investment = $\lambda_s M_s$	$\mathbb{R}_+$
ℓ_s	Lottery stake = $(1 - \lambda_s)M_s$	$\mathbb{R}_+$
b_s	Base income = $\kappa_s + \kappa_s^\gamma - \delta_w$	$\mathbb{R}$
$\bar{b}(\mu)$	Subsistence floor = $\bar{b}_0 R \mu$	$\mathbb{R}_{++}$
$\kappa^{\min}(\mu)$	Min. upskilling; solves $\kappa + \kappa^\gamma = \bar{b} + \delta_w$	$\mathbb{R}_{++}$
$\Psi(\mu)$	Sensitivity = $d\kappa^{\min}/d\mu$	$\mathbb{R}_{++}$
λ_s^*	Optimal fraction = $\kappa^{\min}/M_s$	$(0, 1)$
$a(\mu)$	Worker L stake = $R\mu - \kappa^{\min}$	$\mathbb{R}_{++}$
$c(\mu)$	Worker H stake = $R(1 - \mu) - \kappa^{\min}$	$\mathbb{R}_{++}$
$\Pi(\mu)$	Total pool = $Wa + c$	$\mathbb{R}_{++}$
Σ^2	Squared spread = $c^2 + W^2 a^2$	$\mathbb{R}_{++}$
ϵ_L^*	Net transfer to L = $(c - Wa)/2$	$\mathbb{R}$
K_{ind}^*	Individual upskilling = $2\kappa^{\min}$	$\mathbb{R}_{++}$
K_{agg}^*	Aggregate upskilling = $(W + 1)\kappa^{\min}$	$\mathbb{R}_{++}$
$\tilde{\rho}_s$	Risk premium = $\theta_s \Sigma^2 / 4$	$\mathbb{R}$
V_s	Equilibrium utility = $\bar{b} + \epsilon_s + \tilde{\rho}_s$	$\mathbb{R}$

Firm objects

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A_t	Asset stock at period t	$\mathbb{R}_+$
$\Phi(\mu)$	Upskilling index = $(W + 1)\kappa^{\min}/R$	$\mathbb{R}_{++}$
λ^*	Optimal labor share = $\Gamma/4$	$(0, 1)$